

Embedded Systems Lab Setup & Sensor Interfacing – Task 2

Smart Lighting System Using LDR and Arduino UNO

Name: D.Sushanth

Aim

To design and implement a Smart Lighting System using an LDR sensor and Arduino UNO that automatically controls an LED based on ambient light intensity.

Objective

To understand sensor interfacing with Arduino and develop an automatic lighting system that turns ON during low-light conditions and turns OFF when sufficient light is available.

Working Environment

Platform Used: Tinkercad Circuits

Microcontroller: Arduino UNO

Sensor Used: LDR (Light Dependent Resistor)

Components Required:

- LED
- 220Ω Resistor
- Connecting Wires

Theory

A Light Dependent Resistor (LDR) is a sensor whose resistance changes according to the intensity of light. When light falls on the LDR, its resistance decreases. In darkness, its resistance increases.

The Arduino reads the analog value from the LDR and compares it with a predefined threshold value. If the light intensity is below the threshold, the LED turns ON automatically. If the light intensity is above the threshold, the LED turns OFF.

This system is an example of smart automation used in street lights, homes, and industrial lighting applications.

Circuit Connections

Component	Arduino Connection
LDR Terminal 1	5V
LDR Terminal 2	A0
10kΩ Resistor	Between A0 and GND
LED Positive (Anode)	D13
LED Negative (Cathode)	GND through 220Ω resistor

Arduino Code

// SMART LIGHTING SYSTEM USING LDR AND ARDUINO UNO

// Pin Definitions

const int ldrPin = A0; // LDR connected to Analog Pin A0

const int ledPin = 13; // LED connected to Digital Pin 13

//Threshold Value

int threshold = 500;

// Setup Function

void setup()

{

 // Initialize Serial Monitor

 Serial.begin(9600);

 // Configure LED Pin as Output

 pinMode(ledPin, OUTPUT);

}

// Main Loop Function

void loop()

{

 // Read Light Intensity from LDR

 int lightValue = analogRead(ldrPin);

```
// Display Sensor Value on Serial Monitor
Serial.print("Light Value: ");
Serial.println(lightValue);

// Check Light Intensity
if (lightValue < threshold)
{
    // Low Light Condition
    digitalWrite(ledPin, HIGH);

    Serial.println("LED ON - Low Light Detected");
}
else
{
    // Bright Light Condition
    digitalWrite(ledPin, LOW);

    Serial.println("LED OFF - Sufficient Light Available");
}

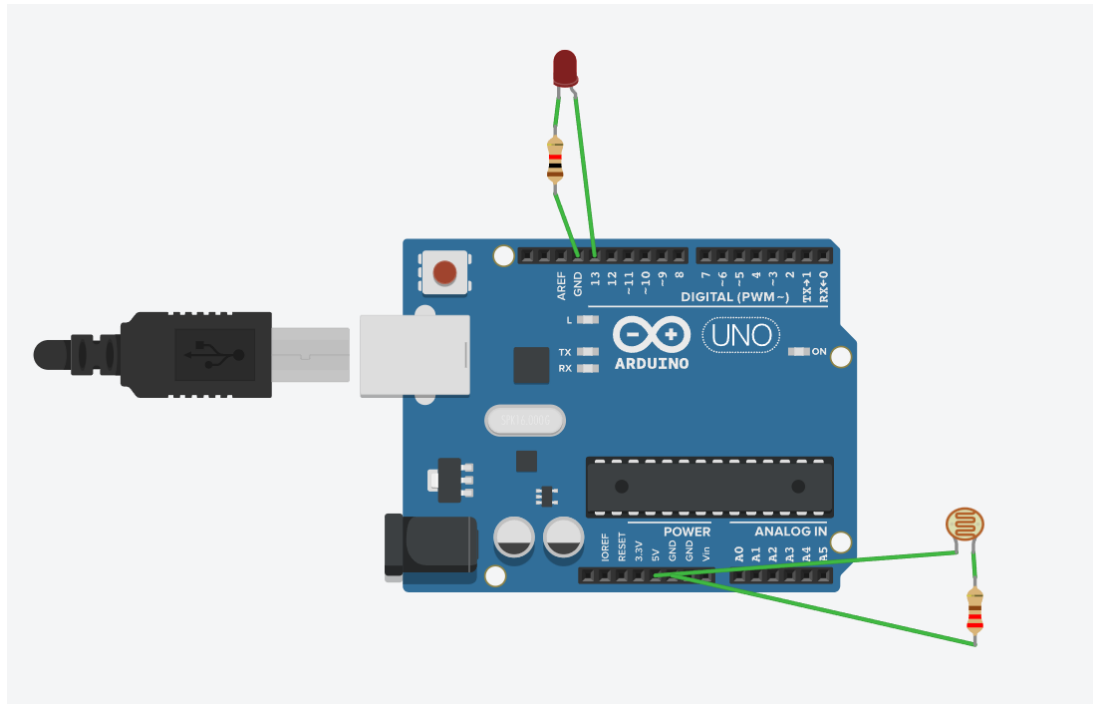
// Delay for Stability
delay(1000);
}
```

Working Procedure

1. Connect the LDR to Arduino pin A0 and 5V.
2. Connect a 10k Ω resistor between A0 and GND.

3. Connect the LED to digital pin D13 through a 220Ω resistor.
4. Upload the code to Arduino.
5. Start the simulation in Tinkercad.
6. Change the light intensity using the LDR slider.
7. Observe the LED and the Serial Monitor output.

Circuit Diagram:



Observation Table

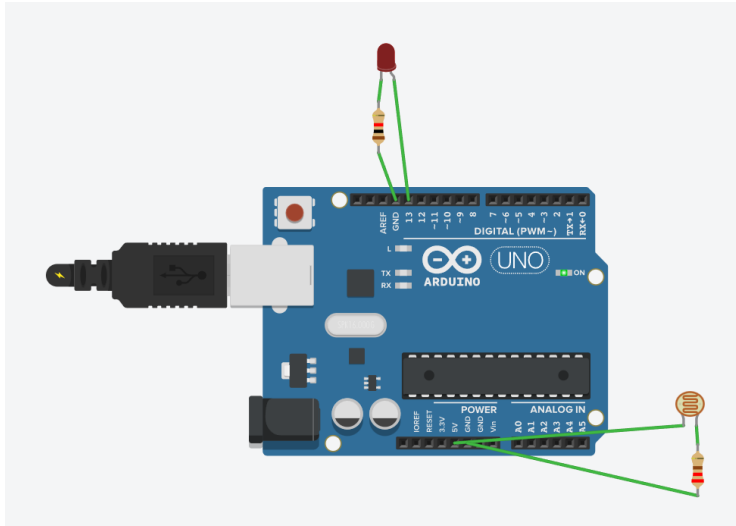
Light Condition Sensor Value LED Status

Bright Light	800	OFF
Normal Light	650	OFF
Low Light	400	ON
Dark Condition	200	ON

Output

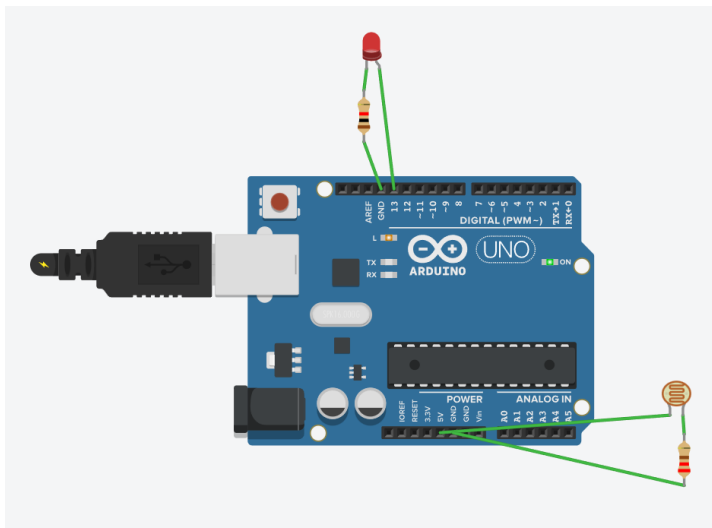
Bright Environment

```
-----
Light Value: 1006
LED OFF - Sufficient Light Available
```



Dark Environment

```
LED ON - Low Light Detected
Light Value: 343
LED ON - Low Light Detected
Light Value: 299
LED ON - Low Light Detected
```



Links:

https://www.tinkercad.com/things/6upr9nNCN7y-smart-lighting-system?sharecode=LJQ3OIRjbCF2x_71C7Ew0FV0-oiy1w4ceLYuWWADets

Applications

- Automatic Street Lighting
- Smart Home Automation
- Energy-Efficient Lighting Systems

- Garden Lighting
- Parking Area Illumination

Result

The Smart Lighting System was successfully implemented using an LDR sensor and Arduino UNO in Tinkercad. The system automatically controlled the LED according to ambient light intensity. The project demonstrated successful sensor interfacing and real-time automation.

Conclusion

The experiment proved that an LDR sensor can be effectively used with Arduino to create a smart lighting system. The system automatically responds to environmental lighting conditions and can be applied in various energy-saving automation projects.